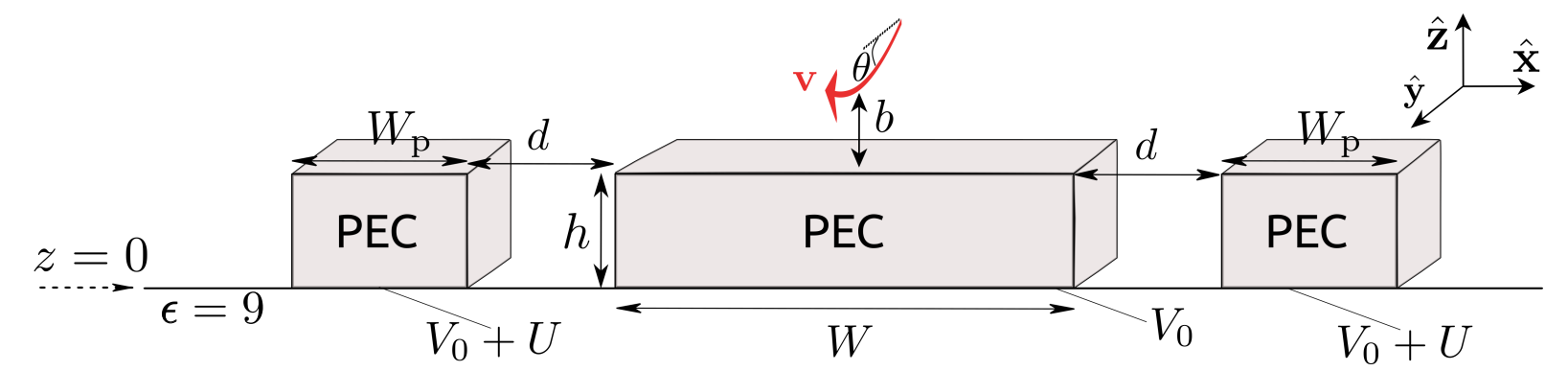


Electron-coupling to waveguide: code of the EELS

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{\substack{b_{\min}+h \\ z_{\min} \\ \text{step 2b}}}^{\infty} \frac{v dz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(x,z)}{m_e \gamma_e} \right|}} \underbrace{\frac{d\Gamma_{\text{EELS}}(\omega, x, z)}{dy}}_{\text{step 1}}$$



1) fit the dGamma/dy with an exponential using ../bem2D : plot_EELS_fitted_exp.py

$$\frac{d\Gamma}{dy}(\omega, z) = A e^{-2\kappa(z-h)},$$

2) run_potential_and_find_bmin.sh :

- a) Run e.out to create normalized potential data
g++ -o e.out e.cpp
files e.cpp and e.h

- b) Run Python script to obtain bmin(V0,theta)
find_bmin_vs_V0_theta.py

solutions of this Eq:

$$\left. \begin{array}{l} \text{b) Run Python script to obtain bmin(V0,theta)} \\ \text{find_bmin_vs_V0_theta.py} \end{array} \right\} 0 = \frac{V(z_{\min})}{U} \frac{U}{V_0} + \frac{m_e c^2 \gamma_e}{2e} \frac{\beta^2 \sin^2 \theta}{V_0} \longrightarrow \theta, V_0 \text{ for a fixed } z_{\min}$$

3) EELS_integrated_over_z.py

$$\Gamma_j(x) \approx 2\hbar\beta\Delta\omega_j A(\omega_j) \cdot \int_{z_{\min}}^{\infty} \frac{dz e^{-2\kappa(\omega_j)(z-h)}}{\sqrt{|2e\phi(x,z)/(m_e c^2 \gamma_e) - \beta^2 \sin^2 \theta|}}, \quad (19)$$